

Engineers Don't Drive Trains—They Pack Them

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Abstract

Given space and weight constraints and a local trucking restriction, two railroad flatcars are to be packed to minimize wasted floor space. This analysis looks at the larger problem as a series of smaller, easy-to-solve problems and determines the set of 30 optimal solutions.

First, the upper bound for maximizing floor space used in the two cars is determined to be 2039.4 cm. Only one set of crates (8 of C1, 7 of C2, 9 of C3, 6 of C4, 3 of C5, and 3 of C6 crates) has a total thickness of 2039.4 cm. The problem becomes that of determining if any feasible packing configuration exists that attains the upper bound. The weight constraint is less restricting than the thickness constraint and can thus be ignored for the time being. As a further simplification, the six different types of crates are put into four groups based on thickness. A list of 6 optimal solutions based on thickness grouping is determined. By splitting up the thickness groupings back into crate types, the list of 6 optimal solutions is expanded into all possible combinations with a 2039.4 cm total thickness. Lastly, the weight constraint limits the number of optimal solutions to 30.

The strengths and weaknesses of the model are then discussed, as are methods of judging the 30 optimal solutions.

1. Introduction

When determining the loading patterns for crates of varying thickness and weights on railroad flatcars, the problem of minimizing wasted floor space arises. The given problem is concerned with

Table 1
Types of Crates.

	Category A				Category B		
	C1	C2	C3	C4	C5	C6	C7
Thickness	48.7	52.0	61.3	72.0	48.7	52.0	64.0 cm
Weight	2000	3000	1000	500	4000	2000	1000 kg
Number	8	7	9	6	6	4	8

loading seven different types of crates (see Table 1) onto two flat railroad cars; each car has 10.2 m of length available for packing the crates and a weight restriction of 40 metric tons. A local truck restriction limits the total thickness of crate types C5, C6, and C7 to 302.7 cm.

2. Assumptions

The analysis is based on the following assumptions:

1. Crates of the same type do not have to be packed on the same car.
2. Empty space does not have to be provided between packed crates.
3. As the two cars are identical, any crate can be assigned to either car.
4. Crates can only be stacked side by side (i.e., there cannot be any other packing method).
5. The widths and heights of the crates do not affect crate packing.
6. The thickness of the crates and the available packing length of the cars are absolutes.
7. It is not important to distribute the weight of the crates evenly between the two cars, as long as the weight restriction is observed.

3. Analysis

3.1. Weight Constraint

After some initial calculations, it is apparent that not all of the crates can be packed in the two railroad cars; the total thickness of all the crates is 2749.5 cm, whereas only 2040 cm is available. Thus, at most 74.2% of the total thickness can be packed; 74.2% of the total weight (89 metric tons) is 66.038 metric tons, half of which (33 metric tons) falls well below the 40 metric tons per car weight

constraint. Therefore, to simplify the problem, the approach taken will focus primarily on the thickness constraint.

3.2. Upper Bound on Thickness

A theoretical upper bound for maximizing floor space used is found by adding the 302.7 cm allowed for crates of type C5, C6, and C7 to the total thicknesses for all of the crates of types C1, C2, C3, and C4 (1737.3 cm). This upper bound is 2040 cm, which is the space available on the two cars. However, meeting this upper bound is contingent upon determining:

- (1) a favorable packing pattern, and
- (2) an integer combination of C5, C6, and C7 that adds to 302.7 cm in total thickness.

3.3. Crates C5, C6 and C7

To narrow the focus of the analysis, only crates of type C5, C6, and C7 are examined, as all of crate types C1, C2, C3, and C4 must be packed to meet the upper bound.

To determine how many crates of types C5, C6, and C7 to pack, an APL program was written to determine combinations of these crates which have a combined thickness between 290 cm and 302.7 cm. It is reasonable to check all the possible combinations, as only 315 exist. A listing of these combinations is found in Table 2. A lower bound of 290 cm was chosen because it is near to 292.2 cm, which is the total thickness of all the crates of type C5, an easily verifiable lower bound.

Three crates of C5 and three crates of C6 have a total thickness of 302.1 cm, which is the total thickness closest to 302.7 cm. So the

Table 2.
Combinations of crates of type C5, C6, and C7 with a combined thickness between 290 cm and 302.7 cm.

Number of crates			Combined Thickness
C5	C6	C7	
3	3	0	302.1 cm
4	2	0	298.8 cm
0	2	3	296.0 cm
5	1	0	295.5 cm
1	1	3	292.7 cm
6	0	0	292.2 cm

revised upper limit for maximizing floor space used is 2039.4 cm ($302.1 + 1737.3$).

3.4. Packing Configurations for One Car

As the optimal set of crates has been found (all of types C1, C2, C3, C4, and three of each of type C5 and type C6), the problem now is to determine whether or not the crates chosen can be packed into the two cars.

Again the problem is simplified. Since thickness, not weight, is our primary concern, the six different types of crates are placed into four groups based on thickness. In other words, crates of type C1 and C5 (both 48.7 cm) are grouped together, as are crates of type C2 and C6 (both 52 cm). This is reasonable, as the crates of different types within the same group can be moved from one car to another to meet the weight constraint without affecting the total thickness of either car.

Instead of trying to pack both cars simultaneously, concern has to be given to only one car. If it is possible to pack one car so that the space occupied is between 1019.4 cm and 1020.0 cm, then the remaining crates could be packed into the other car as their combined thickness would have to fall between 1020.0 cm and 1019.4. In this case, the optimal solution exists.

Another computer program was written to determine if any configurations of the crates exist such that a car could be packed close to capacity. This program generated all possible 9240 packing configurations for all the ways a single car can be packed.

The output from this program shows that configurations do exist that produce an optimal solution (with regard to the thickness constraint—however, as noted above, the weight constraint was not a problem). Therefore, only these configurations have to be examined further.

3.5. Packing Configurations for Both Cars

A third computer program, was written to identify ways in which all of the crates could be packed into both cars. The wasted floor space is only 0.6 cm ($0.6 = 2040 - 2039.4$). The six different configurations found are shown in Table 3. Note, however, that these configurations are not the final solutions, as crates are grouped by length and not by type.

A fourth computer program was written to determine possible packing configurations of the six types of crates based on the six

Table 3.
Ways to split the chosen crates so they fit into the 2 flat cars.
(Note: The crates are grouped by lengths, not by type.)

CAR 1				CAR 2			
No. of crates				No. of crates			
48.7 cm	52 cm	61.3 cm	72 cm	48.7 cm	52 cm	61.3 cm	72 cm
6	0	6	5	5	10	3	1
4	7	4	3	7	3	5	3
6	2	9	1	5	8	0	5
2	0	8	6	9	10	1	0
0	7	6	4	11	3	3	2
11	1	0	6	0	9	9	0

Table 4.
Optimal solutions.

No.		Thickness	Weight	C1	C2	C3	C4	C5	C6
1	CAR 1	1020.0	34.5	4	7	4	3	0	0
	CAR 2	1019.4	32.5	4	0	5	3	3	3
2	CAR 1	1020.0	33.5	4	6	4	3	0	1
	CAR 2	1019.4	33.5	4	1	5	3	3	2
3	CAR 1	1020.0	32.5	4	5	4	3	0	2
	CAR 2	1019.4	34.5	4	2	5	3	3	1
4	CAR 1	1020.0	31.5	4	4	4	3	0	3
	CAR 2	1019.4	35.5	4	3	5	3	3	0
5	CAR 1	1020.0	36.5	3	7	4	3	1	0
	CAR 2	1019.4	30.5	5	0	5	3	2	3
6	CAR 1	1020.0	35.5	3	6	4	3	1	1
	CAR 2	1019.4	31.5	5	1	5	3	2	2
7	CAR 1	1020.0	34.5	3	5	4	3	1	2
	CAR 2	1019.4	32.5	5	2	5	3	2	1
8	CAR 1	1020.0	33.5	3	4	4	3	1	3
	CAR 2	1019.4	33.5	5	3	5	3	2	0
9	CAR 1	1020.0	38.5	2	7	4	3	2	0
	CAR 2	1019.4	28.5	6	0	5	3	1	3
10	CAR 1	1020.0	37.5	2	6	4	3	2	1
	CAR 2	1019.4	29.5	6	1	5	3	1	2
11	CAR 1	1020.0	36.5	2	5	4	3	2	2
	CAR 2	1019.4	30.5	6	2	5	3	1	1
12	CAR 1	1020.0	35.5	2	4	4	3	2	3
	CAR 2	1019.4	31.5	6	3	5	3	1	0
13	CAR 1	1020.0	39.5	1	6	4	3	3	1
	CAR 2	1019.4	27.5	7	1	5	3	0	2
14	CAR 1	1020.0	38.5	1	5	4	3	3	2
	CAR 2	1019.4	28.5	7	2	5	3	0	1
15	CAR 1	1020.0	37.5	1	4	4	3	3	3
	CAR 2	1019.4	29.5	7	3	5	3	0	0

Table 4 (continued)

No.		Thickness	Weight	C1	C2	C3	C4	C5	C6
16	CAR 1	1019.9	27.5	6	2	9	1	0	0
	CAR 2	1019.5	39.5	2	5	0	5	3	3
17	CAR 1	1019.9	29.5	5	2	9	1	1	0
	CAR 2	1019.5	37.5	3	5	0	5	2	3
18	CAR 1	1019.9	28.5	5	1	9	1	1	1
	CAR 2	1019.5	38.5	3	6	0	5	2	2
19	CAR 1	1019.9	27.5	5	0	9	1	1	2
	CAR 2	1019.5	39.5	3	7	0	5	2	1
20	CAR 1	1019.9	31.5	4	2	9	1	2	0
	CAR 2	1019.5	35.5	4	5	0	5	1	3
21	CAR 1	1019.9	30.5	4	1	9	1	2	1
	CAR 2	1019.5	36.5	4	6	0	5	1	2
22	CAR 1	1019.9	29.5	4	0	9	1	2	2
	CAR 2	1019.5	37.5	4	7	0	5	1	1
23	CAR 1	1019.9	33.5	3	2	9	1	3	0
	CAR 2	1019.5	33.5	5	5	0	5	0	3
24	CAR 1	1019.9	32.5	3	1	9	1	3	1
	CAR 2	1019.5	34.5	5	6	0	5	0	2
25	CAR 1	1019.9	31.5	3	0	9	1	3	2
	CAR 2	1019.5	35.5	5	7	0	5	0	1
26	CAR 1	1019.8	29.0	0	7	6	4	0	0
	CAR 2	1019.6	38.0	8	0	3	2	3	3
27	CAR 1	1019.8	28.0	0	6	6	4	0	1
	CAR 2	1019.6	39.0	8	1	3	2	3	2
28	CAR 1	1019.8	27.0	0	5	6	4	0	2
	CAR 2	1019.6	40.0	8	2	3	2	3	1
29	CAR 1	1019.7	34.0	8	1	0	6	3	0
	CAR 2	1019.7	33.0	0	6	9	0	0	3
30	CAR 1	1019.7	33.0	8	0	0	6	3	1
	CAR 2	1019.7	34.0	0	7	9	0	0	2

Note: No crates of type C7 are used.

optimal configurations. For the first time, the weight constraint is taken into consideration. This program identifies the 30 different optimal solutions to the given problem. Table 4 lists these solutions.

4. Testing the Model

Since the feasible upper bound for maximizing floor space used was met, this approach has found the set of optimal solutions. Some calculations were done by hand to check for errors.

5. Strengths and Weaknesses of the Model

The primary strength of the approach is that it decomposes a complex problem into smaller, manageable problems. By solving the small problems one at a time, it is possible to find the optimal solution to the large problem. The approach used is neither complicated nor time-consuming; simple logic is used to determine upper and lower bounds for the solution. Lastly, the optimal set of solutions is found (not just a "close approximation," as may often be found by heuristics; nor just a single optimal solution, found using mathematical programming) by an analysis based on simple assumptions.

While this approach works well with this problem, it would not be as applicable to a problem with more crates of differing thicknesses and a larger number of railroad flatcars. In such a case, the number of different packing combinations would be too large to be solved effectively by this method. Also, this methodology was formulated around the fact that the space available exactly equals the total thickness of crates of types C1, C2, C3, and C4 and the allowable thickness of crates of types C5, C6, and C7. While this is a weakness for solving a problem in which the total thickness of crates of types C1, C2, C3, and C4 is greater than 1737.3 cm, the approach is still extremely simple and effective for a given problem.

Other approaches considered were trim-loss heuristics and linear programming. However, the trim-loss heuristics do not guarantee an optimal solution. The linear programming approach involves rounding the decision variables up or down, which works best for problems of much larger magnitude.

For a more complicated problem (i.e., the weight and thickness constraints are both binding, or a larger number of different crates and cars exist), one of these approaches might be more effective than the one used. Nevertheless, the approach used quickly focused on key issues in the problem; through logical steps, the solution was easily found.

6. Conclusion

The approach used produced 30 different solutions to the packing problem. These solutions could be further evaluated based on additional considerations, such as how the weight is distributed between both cars and if crates of one type are all located on the same car. For example, solutions 29 and 30 could be called the

Table 5.
"Best" solutions.

No.		Thickness	Weight	C1	C2	C3	C4	C5	C6
29	CAR 1	1019.7	34.0	8	1	0	6	3	0
	CAR 2	1019.7	33.0	0	6	9	0	0	3
30	CAR 1	1019.7	33.0	8	0	0	6	3	1
	CAR 2	1019.7	34.0	0	7	9	0	0	2

"best" solutions, as the two cars are carrying approximately equal amounts of weight and only one type of crate is split between both cars (see Table 5).

Appendix A. Description of Computer Programs [omitted]

Appendix B. Program Listings and Output [omitted]

6. Conclusion

The approach was designed to find a solution to the problem. The solution could be found by using the program. The program was designed to find a solution to the problem. The solution could be found by using the program. The program was designed to find a solution to the problem. The solution could be found by using the program.

Appendix C. Problem Analysis— A Flowchart

